“Understanding the Formation of Interest Rates on Credit Loans”
ITESO

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1. Components of Interest Rates:

According to Buckley, the interest rate can be decomposed into five fundamental components, each reflecting a distinct source of risk or economic condition.

1. **Real Risk-Free Rate**
   Constitutes the baseline return required by lenders in the complete absence of risk, as approximated by sovereign debt instruments issued by governments with monetary sovereignty.
2. **Expected Inflation**
   Compensates lenders for the anticipated erosion of purchasing power over the life of the instrument, adjusting the nominal return to preserve the real value of future cash flows.
3. **Default-Risk Premium**
   Represents an incremental spread above the risk-free rate, demanded by investors as compensation for the probability that the borrower may fail to meet its contractual debt obligations.
4. **Liquidity Premium**
   Reflects the additional return required by investors to hold assets that cannot be readily converted into cash at fair market value without incurring significant transaction costs or price concessions.
5. **Maturity Premium**
   Captures the positive relationship between the yield of a fixed-income security and it's time to maturity, compensating investors for the heightened uncertainty and interest rate sensitivity associated with longer-duration instruments.

Data compiled from Day Trading (Buckley, 2023)

2. Interest Rates Across Two Countries:

Countries selected: United States of America and Mexico.

Gathered Data:

Type	USA (%)	Mexico (%)
Personal Loans	12.26	11.22
Mortgages	6.09	6

Data compiled from Yahoo Finance (Nowacki, 2025), Trading Economics (2026), and Mexico Life Realty (2025).

Analysis of Differences:

i. **Economic Factors:** Inflation rates, economic stability, monetary policies.

- Inflation Rates

	USA	Mexico
Inflation Rate	2.9%	4.02%

Data compiled from Federal Reserve (2024) and Trading Economics (2026).

Inflation in the United States measures how fast prices for goods and services increase over time, usually measured using the Consumer Price Index (CPI) or the Personal Consumption Expenditures (PCE) index. The Federal Reserve targets a long-term inflation rate of about 2% to maintain price stability (Federal Reserve, 2024). On the other side, Mexico's inflation rate is measured by the National Consumer Price Index (INPC). The Bank of Mexico aims to keep inflation around 3% (±1%) to maintain price stability in the economy (Banco de Mexico, 2025).

- Economic Stability

  - United States

The U.S. economy in early 2026 exhibits a GDP growing around 1.9%–2%. Second-quarter growth figures released in September indicate a solid expansion, putting the U.S. economy on track for 2 percent annual growth, according to International Monetary Fund estimates (Ferguson & Hippold, 2025). Renter (n.d.) explains that the U.S. economy has remained relatively strong with low unemployment and slowing inflation.

  - Mexico

The Mexican gross domestic product expanded by 0.9% from the previous quarter in the final three months of 2025, rebounding from the 0.3% contraction in the earlier period and revised higher from the preliminary estimate of 0.8% growth to mark the sharpest pace of expansion in over one year (Trading Economics, 2026). According to the International Monetary Fund,

Mexico's economy remains relatively stable but faces challenges such as inflation pressures and slower global demand (International Monetary Fund, 2025).

- Monetary Policies
    - United States

Monetary policy in the United States is managed by the Federal Reserve, which takes actions and communicates policy decisions to support a strong and stable economy. Congress has given the Federal Reserve a dual mandate, which requires it to promote maximum employment and stable prices. Maximum employment refers to the highest level of employment that the economy can sustain without causing excessive inflation, helping create broad economic opportunities for workers. At the same time, the Federal Reserve works to maintain price stability so that consumers and businesses can make long-term financial decisions without worrying about large changes in prices. To achieve this, the Federal Open Market Committee aims for a long-term inflation target of about 2 percent, measured using the Personal Consumption Expenditures (PCE) price index. By adjusting interest rates and influencing the supply of money in the economy, the Federal Reserve helps control inflation, support employment, and maintain overall economic stability in the United States.

    - Mexico

Monetary policy in Mexico is managed by the Bank of Mexico, whose main objective is to maintain price stability and keep inflation low and predictable. The central bank attempts to influence prices through economic signals, using its main policy instrument: the interest rate it charges commercial banks when lending them money. By adjusting this rate, the Bank of Mexico can influence borrowing, spending, and investment in the economy, helping control inflation and maintain overall economic stability. The central bank follows an inflation-targeting framework, aiming for an inflation rate of about 3 percent with a variability range of ±1 percent. To achieve this goal, the Bank of Mexico monitors underlying inflation, analyzes the causes of price changes, and publishes economic reports to guide its decisions. Its primary policy tool is the target interest rate, also known as the interbank interest rate, which influences borrowing, spending, and investment in the economy. By adjusting this rate and communicating its decisions transparently, the Bank of Mexico works to maintain economic stability and control inflation.

ii. **Market Factors:** Supply and demand for credit, competition among lenders.
- Supply and demand for credit

- United States

US dollar corporate credit supply saw a strong start to 2026, with January corporate issuance reaching US$56bn, driven heavily by Technology, Media, and Telecom (TMT) sectors, consistent with high activity in 2025. Total 2025 US corporate supply hit US$926bn, a 14% year-over-year increase, with expectations for continued strong issuance in 2026 (Rahill & Leleux, 2026).

U.S. credit demand reached a record high of $31.8 trillion in early 2025. While corporate bond demand is surging amid high market competition, household debt has also increased, hitting $18.59 trillion in Q3 2024, driven by rising credit card and mortgage, though rejections for new loans remain high.

- Mexico

The supply of credit in Mexico has tightened in recent years, particularly during the COVID-19 pandemic. According to the Bank of Mexico, banks reported stricter lending standards and approval conditions for business loans during 2020. Financial institutions increased collateral requirements, raised the risk premium on loans, and required higher minimum credit scores for small and medium-sized enterprises (SMEs). These measures reflected greater caution from banks due to economic uncertainty and reduced liquidity during the pandemic, contributing to a slowdown in commercial bank lending to businesses (Banco de México, n.d.).

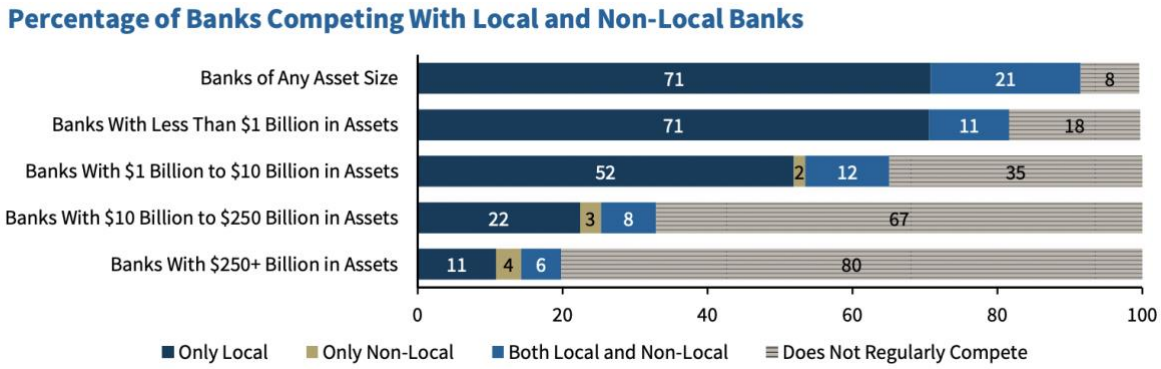
Demand for credit in Mexico also declined during the pandemic period. The Bank of Mexico reported that businesses, especially large companies, initially increased borrowing in early 2020 by using previously established credit lines to compensate for liquidity losses caused by pandemic disruptions. However, demand for new financing decreased significantly afterward as economic activity slowed. Small and medium-sized enterprises had already been experiencing declining credit demand since late 2018, and the pandemic further weakened borrowing activity. Overall, the slowdown in business credit reflected both reduced demand from companies and stricter lending conditions from banks (Banco de México, n.d.).

- Competitions among lenders

  - United States

In the United States, competition among banks remains a significant issue, particularly in the context of continued consolidation within the banking industry. To better understand differences in geographic markets for small business lending, the Federal Deposit Insurance Corporation asked banks to evaluate the importance of several factors that limit the size of their geographic lending markets in its *Small Business Lending Survey 2024: Section 5 — Markets and Competition*. These factors were grouped into three broad categories: competitive factors, operational factors, and location-specific factors. Competitive factors included customer preference for nearby banks, difficulties in forming deposit relationships with small businesses, and challenges in obtaining referrals or identifying potential applicants. Operational factors included difficulties conducting site visits, monitoring loans,

understanding local economic conditions, and verifying applicant information. Location-specific factors referred to insufficient demand outside current markets, a limited number of potential borrowers, and the presence of geographic or political boundaries.
The survey results show that banks in the United States continue to compete primarily with local institutions of similar size in the small business lending market, while competition from nonbank lenders, although growing, remains less dominant in many cases. Nearly two-thirds of surveyed banks reported competing at least sometimes with other financing companies (67%), local banks with $10 billion to $250 billion in assets (64%), and fintech lenders (62%). By contrast, more than half of banks indicated that they rarely or never compete with local banks holding at least $250 billion in assets or with credit card issuers. These findings suggest that despite increasing fintech participation, small business lending competition in the United States remains strongly influenced by local market structures and bank size.



Source: Small Business Lending Survey 2024: Section 5 — Markets and Competition.

In the case of mortgage lending competition in the United States, the market remained highly concentrated in 2024 despite the large number of active lenders. A total of 2,610 lenders originated at least one purchase mortgage during the year, representing a −8.07% decline from the previous year. Although many institutions participated in the market, lending activity was concentrated among a small group of firms: the top 10 lenders funded a combined 26.6% of all purchase mortgages in 2024, while the bottom 4,000 lenders accounted for only 7.7% of total purchase mortgage originations. In addition, just 1.04% of mortgage lenders funded 50.2% of the country's purchase mortgages, highlighting the strong concentration of mortgage lending among a limited number of dominant institutions (Homebuyer.com, 2024).

- Mexico

As reported by Statista in its 2025 analysis *Loan market share of selected banks in Mexico in 2023, by type of loan*, BBVA México concentrated approximately 24% of all loans granted in Mexico in 2023, making it the leading institution in the country's lending market. Its dominance was even stronger in the mortgage segment, where it held a larger market share than in total lending. Banorte ranked second, accounting for almost 15% of total lending, consolidating its position as one of the principal competitors in the Mexican banking system. This market structure reflects the high concentration of credit in Mexico, where the seven largest banks together hold approximately 77% of total credit balances, indicating that lending activity remains heavily concentrated among a limited number of major institutions.

In addition, Banorte was recognized in 2024 as one of the most valuable banking brands in Latin America, reinforcing its importance within the regional financial sector.

In Mexico, consumer credit remains highly concentrated among a small group of large commercial banks, particularly in the credit card segment. According to data from Comisión Nacional Bancaria y de Valores, total credit card lending in the banking system reached 669,664 million pesos in September 2025, with BBVA México holding the largest market share at 33% through a portfolio of 219,009 million pesos. Banamex ranked second with 157,574 million pesos and a 24% market share, followed by Santander with 81,423 million pesos (12%) and Banorte with 71,994 million pesos (11%). Together, these four institutions accounted for approximately 80% of the credit card market, confirming the high concentration of consumer lending within Mexico's banking sector. In annual terms, portfolio growth was strongest in Banorte and Banamex, both with 17% year-over-year growth, while BBVA México expanded by 14%; by contrast, Santander recorded a −1% contraction and HSBC a −4% decline. Credit quality also varied across institutions: the overall system non-performing loan ratio (IMOR) stood at 3.25% in September 2025, while BBVA México registered 2.84%, Banorte 2.88%, Banamex 3.21%, and Santander 3.49%, indicating that the largest banks maintained relatively similar portfolio risk levels despite their dominant market positions.

Cartera de crédito al consumo
Millones de pesos y porcentajes

		Tarjeta de crédito (MDP) 1/			Part. De Mercado	Crecimiento YoY	IMOR (%) 5/		
		sep-24	ago-25	sep-25			sep-24	ago-25	sep-25
	Sistema */	600,392	666,604	669,664	100%	12%	3.47	3.23	3.25
1	BBVA México	192,508	217,245	219,009	33%	14%	3.09	2.64	2.84
2	Santander	82,481	82,812	81,423	12%	-1%	4.06	3.76	3.49
3	Banorte	61,540	70,458	71,994	11%	17%	2.81	2.81	2.88
4	Banamex	134,726	156,264	157,574	24%	17%	3.05	3.24	3.21
5	HSBC	43,387	42,449	41,581	6%	-4%	3.25	4.17	4.17
6	Scotiabank	11,723	13,630	13,755	2%	17%	2.85	2.69	2.73
7	Inbursa	14,874	16,697	16,571	2%	11%	1.23	1.44	1.57
8	Banco del Bajío	2,712	3,050	3,082	0%	14%	2.87	3.11	2.98
9	Banco Azteca	7,805	6,580	6,416	1%	-18%	8.03	7.38	7.38
10	Afirme	1,555	1,769	1,811	0%	16%	3.13	3.50	3.22
11	Banregio	5,906	6,366	6,368	1%	8%	2.88	3.44	3.37
12	Invex	26,029	32,901	33,564	5%	29%	2.13	2.71	2.54
13	BanCoppel	14,624	15,244	15,281	2%	4%	15.31	9.41	9.20
14	**Banca Mifel**	**299**	**430**	**438**	**0%**	**46%**	**6.77**	**5.83**	**5.97**
15	Openbank	-	181	228	0%	0%	-	2.34	4.48
16	Banco Ualá	223	526	571	0%	156%	19.51	9.95	10.04

Source: Comisión Nacional Bancaria y de Valores (CNBV), Cartera de crédito al consumo: Tarjeta de crédito*, septiembre 2025.*

iii. **Regulatory Environment:** Banking regulations, consumer protection laws.

Banking regulations:

- United States

The Federal Reserve promotes a safe, sound, and efficient banking and financial system that supports the growth and stability of the U.S. economy. The Federal Reserve carries out its supervisory and regulatory responsibilities and supporting functions primarily by supervising

the activities of financial institutions to promote their safety and soundness, developing regulatory policy rulemakings, supervision and regulation letters, policy statements, and guidance and acting on applications filed by banking organizations, and monitoring trends in the banking sector by collecting and analyzing data (Federal Reserve System, 2025).
Regulation entails establishing the rules within which financial institutions must operate. This includes issuing specific regulations and guidelines governing the formation, operations, activities, and acquisitions of financial institutions. The Federal Reserve offers numerous resources to assist banking organizations and the public in understanding these rules and related expectations.

U.S. banking laws are based on federal statutes, implementing regulations, and international standards. Key banking laws and regulations include:

- Bank Holding Company Act (BHCA)[vii] – regulates the conduct of BHCs and financial holding companies (FHCs), requiring regulatory approval for acquisitions and expansions.
- Federal Deposit Insurance Act (FDIA)[viii] – establishes deposit insurance protections and outlines regulatory powers of the FDIC over insured banks.
- Dodd-Frank Act and Volcker Rule[ix] – enacted as a reform measure in the wake of the 2008 financial crisis and includes restrictions on proprietary trading and investments in hedge funds and private equity funds under the Volcker Rule.
- BSA[x] – the primary AML/CFT law requiring financial institutions to implement customer due diligence (CDD), suspicious activity reporting (SARs), and risk-based monitoring programmes.
- Community Reinvestment Act (CRA)[xi] – encourages banks to provide credit and services to low-income communities and prevent discriminatory lending practices.
- Equal Credit Opportunity Act (ECOA)[xii] – primary fair lending law that prohibits creditors, including banks, from discriminating against credit applicants on the basis of protected characteristics.

Global Legal Insights, 2025.

- Mexico

Banking regulation in Mexico, supervised primarily by the CNBV and the Banco de México (Banxico), is strict, risk-based, and aligned with international standards under Basel III (CNBV, 2025). The legal framework includes the Credit Institutions Law and anti-money laundering regulations, covering the operation of more than 50 banks, deposit protection through the IPAB, and consumer protection through Condusef. Mexico's financial system is composed of 50 authorized and operating banks, all of which comply with international regulatory standards and are supervised by the CNBV under a risk-based prudential framework (CNBV, 2025).

Mexico's financial system is supervised by **seven main regulators**, each with distinct responsibilities.

- Secretaría de Hacienda y Crédito Público (SHCP): Designs and implements Mexico's financial, fiscal, and public debt policy, and regulates the organization and operation of banks and foreign financial entities.

- Comisión Nacional Bancaria y de Valores (CNBV): Supervises and regulates banks, SOFOMs, and financial technology institutions, ensuring compliance with prudential and operational rules.
- Banco de México (Banxico): Acts as the central bank, responsible for monetary stability, payment systems, and specific banking regulations.
- Instituto para la Protección al Ahorro Bancario (IPAB): Manages deposit insurance and acts as the authority for bank resolution and liquidation.
- Comisión Nacional para la Protección y Defensa de los Usuarios de los Servicios Financieros (CONDUSEF): Protects financial consumers and resolves disputes between users and financial institutions.
- **Comisión Nacional de Seguros y Fianzas (CNSF**): Regulates and supervises insurance and bonding companies.
- Unidad de Inteligencia Financiera (UIF): Monitors financial operations to detect money laundering and terrorism financing risks.

Baker McKenzie, 2024.

Consumer protection laws:

- United States

The Fair Debt Collection Practices Act (FDCPA) is a federal law in the United States that limits the conduct of debt collectors when attempting to recover certain types of debt and protects consumers from abusive, unfair, or deceptive collection practices. Complementing this framework, the Fair Credit Reporting Act (FCRA) regulates how debts and other financial information are reported in consumer credit reports.

Under the FCRA, if a debt collector furnishes information to a credit reporting company that a consumer believes is inaccurate, the consumer has the right to dispute that information. In such cases, credit reporting agencies must indicate on the credit report that the information is disputed, investigate the claim, forward supporting documents to the entity that provided the information, and report the results of the investigation back to the consumer (Consumer Financial Protection Bureau, 2024). In addition, the Consumer Credit Protection Act of 1968 (CCPA) established broader federal protections for consumers in their relationships with banks, credit card companies, and other lenders.

The Act introduced disclosure requirements that must be followed by consumer lenders and auto-leasing firms, including the obligation to disclose the total cost of a loan or credit product, how interest is calculated, and any associated fees. It also prohibits discrimination in lending decisions and bans misleading advertising practices related to credit products (Investopedia, n.d.).

- Mexico

The main statutes governing payment, deposit, and lending services in Mexico include several key legal frameworks. First, the Law to Protect and Defend Financial Services Users (LPDUSF) establishes the protection of users' rights and interests in financial services and defines the

powers of CONDUSEF. Second, the Credit Institutions Law (LIC) regulates banking and credit services, the organization and operation of credit institutions, the activities banks may perform, and the supervisory role of the Mexican state over the banking system. Third, the Transparency and Financial Order Law (LTOSF) regulates fees, exchange rates, and other conditions related to financial services provided by both financial and non-financial institutions. Fourth, the Secured Credit Transparency Law governs financial activities related to secured housing loans, including financing for home acquisition, construction, renovation, and refinancing. Fifth, the General Law of Negotiable Instruments and Credit Transactions regulates negotiable instruments and credit operations, including deposits and lending transactions. Finally, the Financial Technology Institutions Law (LITF) establishes the legal framework for the organization, operation, and supervision of financial technology institutions and innovative financial service models.

Mexico's supervisory framework is led primarily by two institutions. The first is the Comisión Nacional Bancaria y de Valores (CNBV), which has broad regulatory and supervisory authority over most financial institutions, including fintech entities. The CNBV oversees capital requirements, reserve obligations, anti-money laundering measures, know-your-customer policies, and the general operation of credit institutions. The second is the Comisión Nacional para la Protección y Defensa de los Usuarios de Servicios Financieros (CONDUSEF), a decentralized public agency under the Secretaría de Hacienda y Crédito Público (SHCP), whose main role is to protect financial consumers in relation to banks, credit cards, and fintech companies. In addition to handling complaints and defending users' rights, CONDUSEF promotes financial education and transparency so that consumers can make informed decisions regarding financial products offered through both traditional institutions and digital platforms (CONDUSEF, n.d.).

3. **Different Types of Loans Within One Country:**

   1. **Mortgage**

      Mortgage lending in Mexico exhibits several defining structural characteristics. The average loan duration stands at 15 years, with a mean principal amount of 1,372,000 MXN (Condusef, n.d.). With respect to collateral requirements, the financed property itself serves as the underlying security for the obligation, as is standard practice in real estate lending.

      Regarding credit quality, the mortgage default rate remained contained within a narrow band of 3.0% to 3.1% during 2025 (BBVA Research, 2025), suggesting a relatively stable borrower base. In terms of interest rate structure, Mexican mortgage instruments are predominantly governed by fixed-rate agreements, with the applicable rate varying across lending institutions according to their individual pricing policies and risk appetite.

   2. **Credit Card**

      Credit card lending in Mexico represents a significant segment of consumer finance. The repayment cycle operates on a short-term basis, with payment obligations falling due within 20 business days following the statement closing date (BBVA, n.d.). In aggregate, the outstanding credit card portfolio exceeds 380 billion MXN, distributed across more than 27 million active cards, reflecting broad penetration of revolving credit among the Mexican population.

      Unlike mortgage instruments, credit card facilities are predominantly unsecured. In the exceptional cases where collateral is required, lenders typically demand a cash deposit equivalent to the total credit limit or a substantial percentage thereof, effectively converting the product into a secured revolving line.

      The sector's default rate stands at approximately 3.23%, marginally higher than that observed in the mortgage segment, consistent with the unsecured nature of the instrument and the broader risk profile of revolving consumer credit (Juárez. E, 2025).

      With respect to interest rate structure, credit card pricing is determined by a base fixed rate to which the Tasa de Interés Interbancaria de Equilibrio (TIIE) is added. Depending on the specific contractual terms agreed upon at origination, the resulting effective rate may be either fixed or variable over the life of the credit facility (BBVA, n.d.).

Interest Rate Variation Between Loan Types

The divergence in interest rates across mortgage and credit card instruments is fundamentally attributable to differences in risk profile, collateral structure, loan duration, and liquidity considerations; each of which maps directly onto the components of the nominal interest rate discussed previously.

Mortgage loans carry comparatively lower interest rates owing primarily to the presence of real property as collateral. This security arrangement substantially reduces the lender's exposure to default losses, as the underlying asset can be seized and liquidated in the event of borrower insolvency. Consequently, the default risk premium embedded in mortgage rates is relatively contained. Furthermore, the fixed-rate structure prevalent in the Mexican

mortgage market provides lenders with predictable cash flows, reducing the uncertainty premium required over the life of the instrument.

Credit card facilities, by contrast, are predominantly unsecured, meaning that lenders bear the full credit risk of the borrower without recourse to a tangible asset in the event of default. This structural vulnerability necessitates a substantially higher default risk premium, directly elevating the effective interest rate charged to cardholders. The revolving and short-term nature of credit card debt further compounds this dynamic, as borrower financial conditions can deteriorate rapidly within a single billing cycle, limiting the lender's ability to anticipate and mitigate emerging default risk.

Additionally, the composite interest rate structure of credit cards (base fixed rate plus the TIIE) introduces a variable component absent from standard mortgage agreements, reflecting the heightened sensitivity of short-term consumer credit to prevailing monetary conditions.

In sum, the higher interest rates observed in credit card lending relative to mortgages are a rational market response to greater default risk exposure, the absence of collateral, and the structural complexity of revolving credit instruments.

4. **Risk Factors Affecting Interest Rates:**

a. **Risks Considered by Lenders:**

i. **Credit Risk:** Probability of borrower default.

Credit risk is the risk a borrower poses to lenders of failing to pay back money loaned to them. Such a loan default has a negative financial impact on creditors because it halts the flow of cash to them, causing a loss. Therefore, creditors must be careful to assess the credit risk of borrowers before extending them funds. To analyze credit risk, creditors examine and compare a potential customer's income, existing debt load, and their capacity to repay a new loan. They look at any history of debt and payments made as well as collateral that may be associated with their loan, if applicable. The creditor receives interest payments at a rate they set in exchange for taking on a credit risk (Investopedia, n.d.).

ii. **Market Risk:** Changes in market conditions affecting loan value.

Market risk, also known as systematic risk, refers to the uncertainty associated with any investment decision. Price volatility often arises due to unanticipated fluctuations in factors that commonly affect the entire financial market (Corporate Finance Institute, 2020). Systematic risk is not specifically associated with the company or the industry one is invested in; instead, it is dependent on the performance of the entire market. Thus, it is necessary for an investor to keep an eye on various macro variables associated with the financial market, such as inflation, interest rates, the balance of payments situation, fiscal deficits, geopolitical factors, etc.

iii. **Operational Risk:** Risks from internal processes, people, and systems.

Operational risk is a summary of loss resulting from inadequate or failed internal processes, people and systems or from external events. It is one of the key types of risk that businesses and organizations face, alongside strategic risk, credit risk and market risk. Operational risk management (ORM) involves identifying, assessing and mitigating these risks to reduce the likelihood and impact of potential losses (Stryker, 2024).

iv. **Legal and Regulatory Risks:** Changes in laws and regulations.

Legal risk refers to the potential financial loss that may arise due to non-compliance with laws, regulations, or contractual obligations. It includes exposure to lawsuits, fines, and reputational damage (Tamplin, 2023). Regulatory risk is the potential for financial loss due to changes in government regulations or enforcement actions. It can also result from the failure to comply with existing regulations or to adapt to new regulations.

v. **Country Risk:** Economic or political instability in the borrower's country.

Country risk refers to the uncertainty that investing in a particular country may lead to losses due to factors like political instability, economic weakness, currency fluctuations, or government default (Investopedia, n.d.). This risk is especially pronounced in developing

countries, where volatility can affect corporate securities and sovereign debt alike. Investors often analyze political and economic indicators, credit ratings, and global benchmarks, such as comparison to U.S. risk levels, to gauge exposure and may hedge through diversification or financial instruments.

b. Impact on interest rates:

i. **Credit Risk**

High credit risk often results in higher interest rates as lenders seek compensation for the increased risk of default. Borrowers with high creditworthiness, by contrast, benefit from lower rates (Investopedia, n.d.). Credit risk directly affects interest rates because banks incorporate expected and realized credit losses into loan pricing to preserve profitability across the business cycle.

Evidence from the Federal Reserve System shows that banks with higher realized total credit losses also report higher interest and fee income over time, indicating that lenders charge more when credit risk increases. Specifically, a 1% increase in average net charge-offs is associated with a 0.6% increase in average interest and fee income, suggesting that higher interest rates and fees compensate for part of the losses generated by borrower default risk at the loan, regional, and bank levels. In addition to expected credit risk, loan pricing is also influenced by other factors such as bank profitability, risk aversion, supervision and capital regulation, policy uncertainty, and the degree of competition within the industry, all of which shape how aggressively banks adjust lending rates in response to risk conditions (Abdymomunov et al., 2025).

ii. **Market Risk**

Market risk may arise due to changes to interest rates, exchange rates, geopolitical events, or recessions. Market risk refers to the potential financial losses that arise from fluctuations in the prices of financial instruments due to changes in market conditions. These fluctuations can result from various factors, including changes in interest rates, stock prices, commodity prices, and foreign exchange rates (Corporate Finance Institute, 2020). When interest rates rise, the market value of existing bonds typically falls, leading to potential losses for bondholders. Conversely, when interest rates decline, bond prices increase.

iii. **Operational Risk**

Operational risk, arising from inadequate internal processes, systems failures, or human error, indirectly affects interest rates because losses generated by operational disruptions can weaken a financial institution's capital position, increase regulatory scrutiny, and force liquidity-preserving actions that raise the cost of lending. When operational risk increases, banks often respond by tightening credit conditions or incorporating higher risk premiums into loan pricing to offset potential losses. This relationship has become more relevant as the

financial sector faces a more complex operating environment. The Office of the Comptroller of the Currency identified credit risk, market risk, operational risk, and compliance risk as the main risk themes affecting banks, emphasizing that operational risk remains elevated across the industry. According to the OCC, cyber threats continue to intensify as malicious actors target financial institutions and critical service providers through ransomware and other attacks. At the same time, greater digitalization, adoption of innovative financial products, and increased reliance on third parties have made banking operations more complex, generating both opportunities and additional vulnerabilities. Continued check fraud, wire transfer fraud, and rising payment fraud incidents further reinforce the importance of strong fraud risk management, since these operational pressures can increase costs and ultimately influence lending conditions and interest rate decisions (Santika et al.; Office of the Comptroller of the Currency, 2024).

iv. **Legal and Regulatory Risks**

Legal and regulatory risk arises when financial institutions fail to comply with laws, supervisory requirements, or regulatory standards, and its consequences can be severe for both financial stability and lending behavior. Non-compliance often results in substantial financial penalties, which can weaken an institution's capital position and increase operating costs. These additional costs may affect interest rates because banks frequently incorporate higher compliance expenses and risk premiums into loan pricing in order to preserve profitability. In addition to direct financial penalties, non-compliance can damage institutional reputation, reducing customer and stakeholder confidence.

Reputational deterioration may lead to the loss of clients, reduced market share, and weaker access to funding, all of which can further increase the cost of capital and place upward pressure on lending rates. Legal and regulatory failures also generate operational disruptions, including costly audits, corrective measures, and remedial actions that divert resources away from growth strategies and credit expansion. As a result, higher legal and regulatory exposure can indirectly contribute to tighter credit conditions and higher interest rates for borrowers (Riskify, 2025).

v. **Country Risk**

Country risk, which comprises political instability, economic volatility, and sovereign default risk, directly increases interest rates by raising the risk premium demanded by investors, particularly in emerging markets.Some international organizations evaluate the country risk on behalf of their member nations. For example, the Organisation for Economic Co-Operation and Development (OECD), as part of its arrangement regarding officially supported export credits, publishes an updated list of countries and their associated risks for the purpose of setting interest rates and payment terms (Scott, 2025).

5. Case Study Analysis:

The End of Japan's Negative Interest Rate Era

i. What happened?

For nearly three decades, Japan pursued one of the most unconventional monetary policy regimes in modern economic history. Following the asset bubble collapse of the early 1990s, the Bank of Japan (BoJ) progressively dismantled conventional rate policy, introducing the zero-interest rate policy in 1999 and later, in February 2016, cutting its policy rate to –0.1%, making Japan one of the few economies in the world to operate under a formally negative interest rate regime. This framework remained in place for eight consecutive years, becoming a defining feature of Japan's macroeconomic landscape.

The turning point came in March 2024, when the BoJ announced its first policy rate increase since 2007, raising the short-term interest rate from –0.1% to a target range of 0% to 0.1%. This decision marked the formal end of the negative interest rate policy and represented a watershed moment in Japanese monetary history. The BoJ simultaneously dismantled its yield curve control framework, a mechanism introduced in 2016 that had artificially capped 10-year Japanese Government Bond yields at around 0%, further signaling a decisive shift in the central bank's policy orientation.

The tightening cycle did not stop there. In July 2024, the BoJ executed a second rate increase, bringing the policy rate to 0.25%, a level not seen in Japan since 2008. A third consecutive hike followed in January 2025, raising the rate further to 0.5%, marking the highest policy rate in Japan in approximately 17 years. In aggregate, over the course of roughly ten months, the Bank of Japan transitioned from a negative rate environment to a positive policy rate of 0.5%, representing a cumulative increase of 60 basis points from its historic low.

The implications for lending rates were immediate and tangible. Japanese commercial banks, which had long maintained variable mortgage rates near historical lows (in many cases below 0.5%) began announcing upward revisions to their prime lending rates. Major institutions such as Mitsubishi UFJ Financial Group (MUFG), Sumitomo Mitsui Banking Corporation (SMBC), and Mizuho revised their short-term prime rates for the first time in 17 years following the March 2024 decision. Variable-rate mortgage holders, who represent the majority of mortgage borrowers in Japan, faced rising monthly obligations, while fixed-rate products were repriced upward in anticipation of further tightening.

In sum, Japan's interest rate landscape underwent a fundamental transformation between 2024 and early 2025, transitioning from the deepest accommodative stance in its postwar history to a regime of gradual but deliberate monetary normalization.

ii. Why it happened?

The Bank of Japan's decision to exit its negative interest rate policy was not the product of a single triggering event, but rather the convergence of several structural and cyclical developments that, taken together, provided the BoJ with sufficient confidence that

sustainable price stability had finally been achieved. Three interconnected factors explain the shift: the persistence of above-target inflation, a historic breakthrough in wage growth, and mounting pressure from yen depreciation.

a. Persistent Above-Target Inflation

The most fundamental precondition for the BoJ's policy reversal was the sustained breach of its 2% inflation target. Inflation had run above the BoJ's 2% target for 44 consecutive months, with consumer price growth reaching 2.9% in November 2024 (CNBC, 2025). Critically, the BoJ distinguished between inflation driven by transitory cost-push factors, namely, the pass-through of elevated import prices following the pandemic and the energy shock, and inflation that was genuinely demand-driven and domestically anchored. The BoJ projected that the year-on-year rate of increase in the CPI would remain above 2% through fiscal year 2024, partly due to the waning effects of the pass-through of past cost increases and government economic measures (Bank of Japan, 2024). The central bank's stated objective was to confirm that inflation was being sustained by domestic demand dynamics rather than external price pressures; a distinction that proved pivotal in timing the rate decisions.

b. The Wage-Price Virtuous Cycle: The Shunto Breakthrough

Perhaps the most decisive factor in the BoJ's deliberations was the behavior of wages. The Bank of Japan had long conditioned its policy normalization on evidence of a self-reinforcing cycle in which rising wages would support consumer spending, which in turn would sustain price growth, a dynamic it publicly referred to as the "virtuous cycle between wages and prices." The 2024 Shunto (Japan's annual spring labor-management wage negotiations) produced the highest wage increase in 30 years, with the Japanese Trade Union Confederation (Rengo) reporting an overall wage increase of 3.58% (JCER, 2024). More specifically, the initial data indicated a base pay rise of 3.7% and a headline wage increase, including scheduled wage increments, of 5.3%, well above the previous year's final figures of 2.1% and 3.6%, respectively, and representing the first time base pay rose above the 3% threshold since 1991 (Goldman Sachs, 2024). The significance of this result cannot be overstated: base pay, as distinct from seniority-based increments, reflects a structural upward revision of the wage schedule itself, and is therefore the indicator most directly relevant to sustaining inflation over the medium term. The BoJ's own assessment was that, given improving corporate profits and tight labor market conditions, it was highly likely that wages would continue to increase steadily, following the firm increases already observed (Bank of Japan, 2024).

c. Yen Depreciation and Imported Inflation

A third, compounding driver was the sustained depreciation of the Japanese yen, which introduced an additional inflationary dimension that complicated the BoJ's stance. The yen had weakened significantly against the U.S. dollar and other major currencies, raising the cost of imported food, fuel, and other essential inputs for the world's fourth-largest economy (CommBank, 2025). This import cost inflation was further amplified by global capital flows: the strong appetite for investing in dollar-denominated assets linked to the artificial intelligence boom had also pulled capital out of the yen and into dollars (CommBank, 2025).

The resulting dynamic created a situation in which the BoJ needed to stabilize the yen and manage inflationary pressures simultaneously, as yen depreciation was raising concerns about increased import costs and reduced purchasing power (CFI Trades, 2025). By raising borrowing costs, monetary tightening served a dual purpose: anchoring domestic inflation expectations while also providing a degree of support to the exchange rate.

d. The Policy Reassessment: From Accommodation to Normalization

Taken together, these three factors catalyzed a fundamental reassessment of the appropriateness of the negative rate framework. With both conditions in place (strong wage growth and strong services inflation) the BoJ's Policy Board agreed to exit its negative interest rate policy at its March 2024 meeting (EFG International, 2024). The BoJ simultaneously dismantled yield curve control, signaling that the entire architecture of its unconventional policy era was no longer warranted. Subsequent rate decisions in July 2024 and January 2025 were driven by the same underlying logic: the BoJ had become more confident in the inflation outlook, noting that with wages continuing to rise, underlying CPI inflation had been increasing gradually toward the 2% target (EFG International, 2024). Importantly, the pace of tightening was deliberately calibrated to be gradual, as real interest rates remained deeply negative throughout the cycle, meaning that even at a policy rate of 0.5%, monetary conditions in Japan continued to be broadly accommodative.

iii. Impact on interest rates

a. Short-Term Prime Rates and Variable Mortgage Rates

The most immediate and socially significant impact of the BoJ's tightening was felt in the retail lending market, particularly in the residential mortgage segment. Following the July 2024 rate hike, Mitsubishi UFJ Financial Group announced plans to increase its short-term prime rate for the first time in 17 years, with other major lenders expected to follow, a consequential move given that approximately 75% of personal mortgages in Japan are floating-rate loans directly tied to the short-term prime rate. The mechanics of the adjustment were concrete: major city banks raised the base interest rate for variable mortgages by 0.15 percentage points, from 2.475% to 2.625% in October 2024 (Mizuho n.d.), and while some lenders expanded preferential discounts for new borrowers to absorb part of the increase, existing borrowers bore the full impact since their discounts had been fixed at origination. At the household level, the consequences were measurable: analysis by Mizuho Research estimated that each additional 0.25 percentage point hike would produce an average net annual impact of approximately ¥15,000 per debt holding household, with increased mortgage payments of ¥28,000 significantly outpacing the positive effect from higher deposit income of ¥13,000 (Mizuho n.d.).

b. Fixed-Rate and Long-Term Lending Products

Beyond variable-rate products, the normalization cycle had pronounced effects on fixed-rate mortgages and longer-term corporate lending, both of which are priced in reference to long-term JGB yields rather than the overnight policy rate. The dismantling of yield curve control in March 2024 removed the artificial ceiling that had held 10-year Japanese Government

Bonds yields near 0%, allowing long-term rates to reprice freely in response to market expectations. Following the December 2025 rate decision, the yield on 10-year JGBs rose approximately 5 basis points to 2.019%, while the 20-year JGB yield climbed to 2.975%, both reaching their highest levels since 1999 (CNBC, 2025), a structural shift that translated directly into higher fixed mortgage rates, given that 10-year JGB yields had nearly doubled from approximately 1.1% at the start of 2025 to just under 2.0% by late 2025 (Fisher Investments, 2025). For borrowers opting for fixed-rate products, including the widely-used Flat 35 government-backed mortgage scheme, this implied a durable upward shift in the cost of long-term credit independent of the BoJ's incremental policy adjustments.

c. Corporate Lending and Banking Sector Dynamics

The normalization cycle also restructured the economics of corporate lending in a manner that produced both beneficiaries and casualties. For Japanese banks, the steepening yield curve created a more favorable net interest margin environment for the first time in over a generation, with total bank loans rising 4.5% year-on-year in November 2024, their fastest growth rate in at least 25 years outside of emergency pandemic lending (Fisher Investments, 2025). The flip side, however, was the increased pressure on highly leveraged or low-profitability firms that had relied on near-zero borrowing costs to remain operational: Tokyo Shoko Research estimated the existence of approximately 565,000 so-called zombie firms struggling to service debts from operating profits alone, with a 0.1 percentage point increase in rates projected to expand that figure by roughly 12%, to approximately 632,000 firms (Finimize, 2024), introducing a significant credit risk dimension to the normalization cycle.

d. Sovereign Debt and Fiscal Implications

A final dimension of the rate impact pertained to Japan's sovereign debt dynamics. Given that Japan's debt-to-GDP ratio stood at approximately 230%, the highest among major economies according to IMF data (CNBC, 2025), the rise in JGB yields carried substantial fiscal implications that extended well beyond the private sector. Debt-servicing costs had already exceeded ¥25 trillion in fiscal year 2023, with the finance ministry projecting an increase to ¥27 trillion in fiscal 2024, representing approximately a quarter of the government's annual budget (Finimize, 2024), meaning that each incremental increase in long-term yields translated directly into higher refinancing costs on maturing debt issuances, compounding the structural pressures already bearing on Japan's public finances and raising broader questions about fiscal sustainability over the medium term.

6. Summarize Findings:

This report examines the formation of interest rates on credit loans through five interconnected analytical dimensions: the theoretical decomposition of interest rate components, a comparative cross-country analysis, a segmented analysis of loan types within a single market, an assessment of the risk factors that shape lender pricing decisions, and an empirical case study grounded in a real-world monetary policy event. The objective is to construct a comprehensive and internally consistent understanding of how interest rates emerge from the interaction of macroeconomic conditions, institutional frameworks, market structures, and risk considerations providing a foundation for applied analysis in the field of financial engineering.

a. Components of Interest Rates

The theoretical foundation of this report rests on Buckley's (2023) decomposition of the nominal interest rate into five fundamental components: the real risk-free rate, expected inflation, the default risk premium, the liquidity premium, and the maturity premium. Each component reflects a distinct economic condition or risk dimension that lenders must price in order to preserve the real value of their expected returns. This framework is not merely academic; it serves as the structural backbone through which all subsequent findings can be interpreted. The default risk premium, for instance, explains the pricing differential between secured and unsecured products, while the expected inflation component directly links monetary policy decisions to lending rate outcomes. Understanding these components is essential because it allows the analyst to decompose any observed interest rate into its constituent causes rather than treating it as an opaque market output.

b. Interest Rates Across Two Countries: United States and Mexico

The cross-country comparison between the United States and Mexico reveals that similar nominal lending rates can emerge from structurally different macroeconomic environments. Personal loan rates stood at 12.26% in the United States and 11.22% in Mexico, while mortgage rates were nearly identical at 6.09% and 6.00% respectively. Despite this apparent convergence, the underlying drivers diverge considerably. Mexico's inflation rate of 4.02% exceeded the U.S. rate of 2.9%, placing the Banco de México's inflation-targeting framework under greater pressure and requiring a higher nominal rate to preserve lender purchasing power. On the supply and demand side, U.S. credit markets demonstrated significantly greater depth, with total corporate supply reaching $926 billion in 2025 and household debt exceeding $18.59 trillion, reflecting a more mature and competitive lending environment. In Mexico, credit markets remained more concentrated, with the seven largest banks controlling approximately 77% of total credit balances, which reduces competitive pressure on rate-setting and supports structurally higher spreads. Regulatory frameworks also differed in scope: U.S. banking regulation operates under a complex federal architecture including the Dodd-Frank Act and the Equal Credit Opportunity Act, while Mexico's system is coordinated across seven distinct supervisory bodies, including the CNBV, Banxico, and CONDUSEF, under a risk-based prudential framework aligned with Basel III standards.

c. Loan Types Within Mexico: Mortgages and Credit Cards

The analysis of loan types within Mexico illustrates how the components of interest rates manifest differently depending on the structural characteristics of each credit instrument. Mortgage loans, with an average principal of 1,372,000 MXN and a 15-year duration, carry comparatively lower rates owing primarily to the presence of real property collateral, which substantially reduces the lender's default risk exposure. The mortgage default rate remained contained between 3.0% and 3.1% during 2025, consistent with the lower risk profile of a secured instrument. Credit cards, by contrast, are predominantly unsecured, carry a revolving short-term structure, and exhibit a slightly higher default rate of approximately 3.23%. Their composite pricing mechanism (a fixed base rate plus the TIIE) introduces a variable component that reflects the heightened sensitivity of short-term consumer credit to prevailing monetary conditions. The divergence in rates between these two products is therefore not arbitrary but maps directly onto the Buckley framework: higher default risk premiums and the absence of collateral in credit cards justify the observed rate differential relative to mortgage instruments.

d. Risk Factors Affecting Interest Rates

Lenders incorporate five categories of risk into their pricing decisions: credit risk, market risk, operational risk, legal and regulatory risk, and country risk. Of these, credit risk is the most direct determinant of the interest rate spread above the risk-free rate. Evidence from the Federal Reserve shows that a 1% increase in net charge-offs is associated with a 0.6% increase in average interest and fee income, confirming that loan pricing responds systematically to realized default losses (Abdymomunov et al., 2025). Market risk introduces rate sensitivity through fluctuations in benchmark rates, exchange rates, and asset prices, while operational risk, increasingly amplified by cybersecurity threats and digitalization, indirectly raises lending costs by weakening institutional capital positions. Legal and regulatory risk generates compliance costs that are partially passed through to borrowers via higher rate spreads. Finally, country risk, evaluated through frameworks such as the OECD's export credit classification, elevates the risk premium demanded on sovereign and corporate instruments issued in politically or economically unstable environments.

e. Case Study: Japan's Monetary Normalization (2024–2025)

The Japan case study provides a concrete, real-world illustration of how macroeconomic developments, policy reassessments, and risk repricing interact to produce measurable changes in lending rates across an entire economy. Between March 2024 and January 2025, the Bank of Japan raised its policy rate by a cumulative 60 basis points, from –0.1% to 0.5%, ending an eight-year experiment with negative interest rates. This shift was driven by three converging factors: inflation exceeding the 2% target for 44 consecutive months, a historic breakthrough in the Shunto wage negotiations that produced the highest base pay increase since 1991 at 3.7% (Goldman Sachs, 2024), and sustained yen depreciation that intensified imported inflationary pressures. The transmission to lending rates was both immediate and structurally significant. Major banks raised their short-term prime rates for the first time in 17 years, directly increasing variable mortgage costs for approximately 75% of Japanese mortgage holders. Long-term rates repriced as well, with 10-year JGB yields rising to 2.019% by late 2025, nearly doubling from 1.1% at the start of the year (Fisher Investments, 2025),

driving fixed mortgage rates upward. Corporate lending expanded at its fastest pace in 25 years, yet the normalization simultaneously threatened approximately 565,000 highly leveraged zombie firms whose viability depended on near-zero borrowing costs (Finimize, 2024). At the sovereign level, debt-servicing costs were projected to reach ¥27 trillion in fiscal 2024 against a debt-to-GDP ratio of approximately 230%, underscoring the fiscal dimension of rate normalization in a highly indebted economy.

f. Key Takeaways and Implications for Financial Engineering

Several overarching insights emerge from this analysis. First, interest rates are not exogenously determined prices but endogenous outcomes of the interaction between monetary policy, inflation expectations, credit risk, and institutional market structure, each of which maps onto a specific component of the Buckley decomposition. Second, the cross-country comparison demonstrates that structurally similar nominal rates can conceal fundamentally different risk compositions; financial engineers and credit analysts must therefore disaggregate observed rates into their underlying drivers rather than relying on surface-level comparisons. Third, the segmented loan analysis confirms that collateral, duration, and repayment structure are primary determinants of rate differentiation within a single market, a principle with direct implications for loan pricing models, credit scorecard design, and risk-adjusted return calculations. Fourth, and perhaps most consequentially, the Japan case study illustrates that prolonged policy distortions, such as artificially suppressed rates under negative interest rate and yield curve control regimes, produce structural dependencies in borrower behavior and corporate solvency that significantly complicate the normalization process. The abrupt repricing of credit costs upon exit from such regimes generates first-order effects on household finances, corporate viability, and sovereign debt sustainability simultaneously. For practitioners in financial engineering, this underscores the importance of stress-testing credit portfolios against rate normalization scenarios, particularly in environments where a significant share of borrowers hold floating-rate obligations.

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